Mittuniversitetet
MID SWEDEN UNIVERSITY

DT081A – Signal and Image Processing

Prof Mårten Sjöström & Dr Roger Olsson



Begränsad delning

Visual AI – Motivation

Visual AI Requested from Industry

A recent survey showed the need of

30 000 experts in Sweden in the next 5 years

Experts in
Artificial Intelligence, Machine and Deep Learning

Experts in
Visual Data

Experts with Competence in
Mathematics

Begränsad delning

Course Information

Signal and Image Processing – Course Aims

- To provide a deep understanding of both fundamental and advanced methods within signal and image processing

- To equip students with the theoretical knowledge and practical skills required to successfully apply these methods in areas such as

  - computer vision, image analysis, multidimensional visual representation and compression, advanced image processing, as well as

  - applications within machine learning for visual media.

Signal and Image Processing – Learning outcomes

Upon completing the course, the student will be able:

- To describe and explain the theoretical fundamental principles within signal and image processing.
- To apply the fundamental principles of signal and image processing in both traditional methods and modern AI-based systems.
- To select and apply appropriate signal and image analysis techniques on specific problems with visual media.
- To describe and explain algorithms within signal and image processing based on mathematical foundations.
- To design and implement filters for signals and images in both time, space, and frequency domains.
- To evaluate and critically analyze the performances of different signal and image processing algorithms, using tools and programming languages relevant to the area.

Signal and Image Processing – Course Contents

- Lecture 1 – 3 : Introduction to signal and image processing
  - Signal and system representations, (time/frequency/space), sampling, quantization, image formats, convolution, Hilbert spaces,transforms, quality metrics, and applications.
- Lecture 4 – 5: Time-domain analysis and digital filter design
  - FIR/IIR, impulse response, computational, complexity.
- Lecture 6 – 7: Frequency-domain analysis and filter design
  - Fourier analysis, LP/HP/BP/BS, spectrum, spectrogram.
- Lecture 8 – 10: Spatial-domain analysis
  - filter kernels, spatial filters for sharpening, noise reduction,histograms, contrast enhancement, color space.
- Lecture 11 – 12: Introduction to how signal and image processing are applied in visual AI

- Laboratory exercises: Implementation examples for real-world problems in signal and image processing, as well as analysis of scientific articles.

Signal and Image Processing – Course Prerequisites

- Computer Engineering, 60 credits
  - including object-oriented programming, 15 credits;

- Mathematical subjects, 30 credits
  - including probability theory and statistics, and linear algebra.

Signal and Image Processing – Course Literature and Resources

Resources

- Moodle
  - Lecture notes,
  - Literature references,
  - Exercises,
  - Labs
- Library
- Peers

Signal and Image Processing – Teachers

- Prof. Mårten Sjöström
  Marten.Sjostrom@miun.se, room L418
  - Examiner, lecturer
- Dr. Roger Olsson
  Roger.Olsson@miun.se, room L420
  - Lecturer

<u>Interactions with students</u>

- Mainly during lectures, exercises and labs
- Personal meetings after agreements
  - Book via email

Course Structure

According to Course timetable

Lectures

- Theoretical presentations – Questions by the students (no silence!)

Exercises

- Application of theory – Interaction between teachers and students

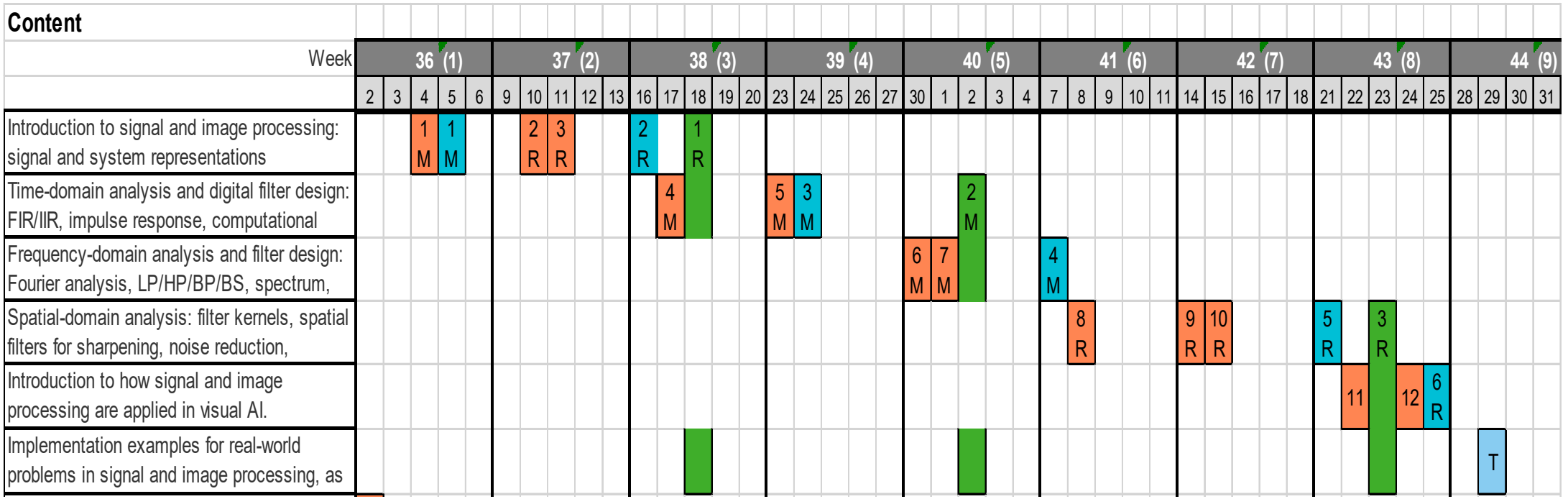
Laboratories

- Practical application – Interaction between students with support by teachers
- Writing lab reports

Examination

- Written examination 4.5 ECTS (A – F) – 3 times per year
- Lab reports 1.5 ECTS (P / F) – corrected 3 times per year (hand in immediately!)

Course Planning



- Course time table online

Advanced course – You are Master Students!

- More profound knowledge and specialization
  - Deeper and more specialist knowledge, applying fundamental knowledge from bachelor courses.
- More independent studies
  - You drive your own learning process: planning and execution, identifying required knowledge and where to find it.
- More emphasis on critical thinking
  - Higher degree of analysis, synthesis, evaluate research articles, and contribute with own results.
- Greater responsibility and involvement
  - Own responsibility for your learning: more advanced knowledge and projects; collaborations with peers in projects.
- Advanced methodology
  - Learn and apply research methodology to produce credible results in own studies.
- Master thesis
  - Larger, independent profound research and development project.

Advanced course – You are Master Students

- 6 ECTS = 160 hours of study hours – at least!
  - Take the habit of studying 8 hours a day

Approach

- Visit all lectures, exercises, labs
  - Absence at your own risk
- Prepare – execute – report
- Repeat, repeat, repeat!
- Collaborate – Help each other!

Begränsad delning

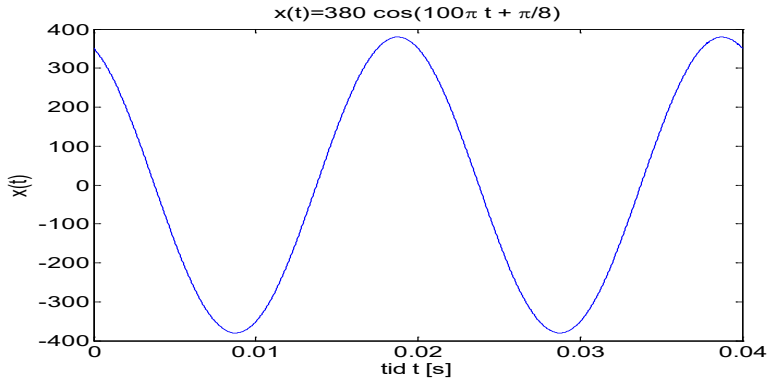
Course Introduction

What is a Signal?

- Signal (generic) = Carrier of information, often changing over time
  - A continuous (1D) function: $f: \Omega \to \mathbb{R}$ $\quad \Omega = \{t \mid a < t < b\} \subseteq \mathbb{R}$

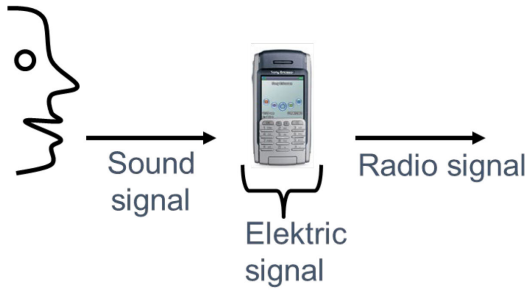
$$t \mapsto y$$

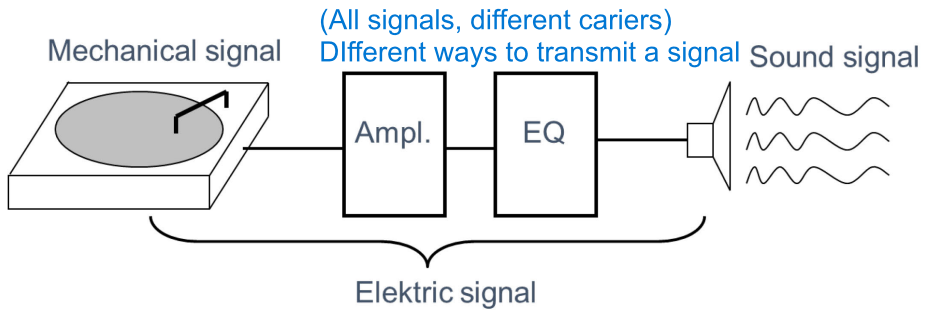
$$\text{or } s = f(t) \in \mathbb{R} \quad t \in (a, b) \subseteq \mathbb{R}$$



SIGNAL OVERVIEW ----------

- Examples
  - Telephony
  - Sound system

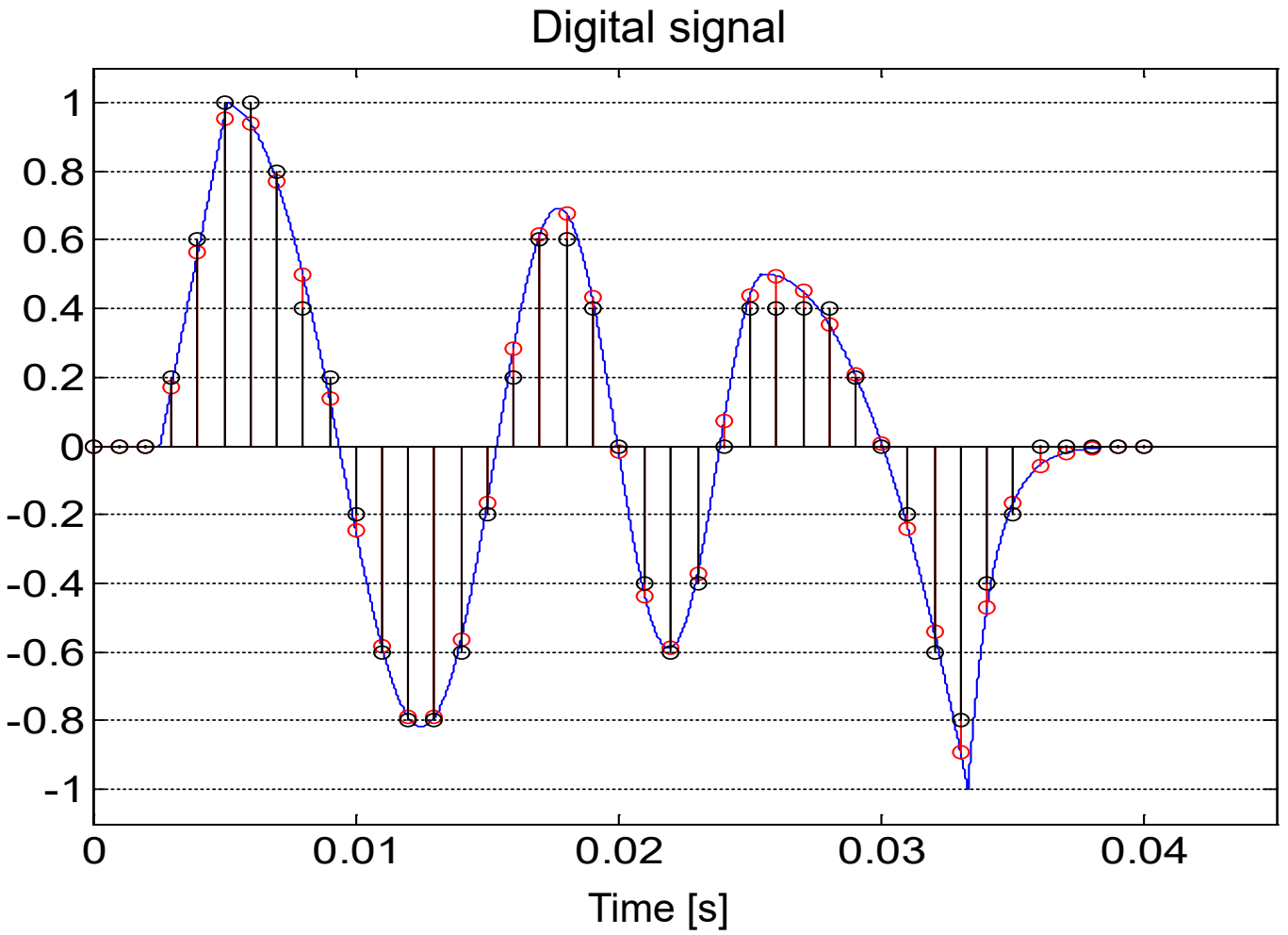




What is a Signal?

- A time-discrete Signal (= sequence of integers) Has no distance between the sgianls
  - A time-discrete function: $f: \Omega \to \mathbb{R}$ $\Omega = \{n \,|a \leq n \leq b\} \subseteq \mathbb{Z}$
    $s = f(n) \in \mathbb{R}$ $n \in [a, b] \subseteq \mathbb{Z}$
  - A digital signal: $f: \Omega \to \mathbb{Z}$ $\Omega \subseteq \mathbb{Z}$
    - Discrete time
    - Discrete amplitue

What is a Signal?



$f: \Omega \rightarrow \mathbb{Z}$

$\Omega \in \mathbb{Z}$

What is Signal Processing?

Filtering

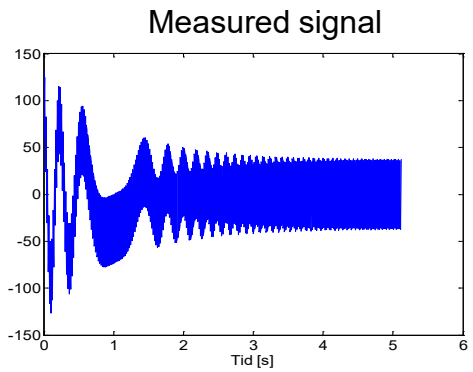
- Accentuating or removing certain aspects = “improving”
- Described as a function

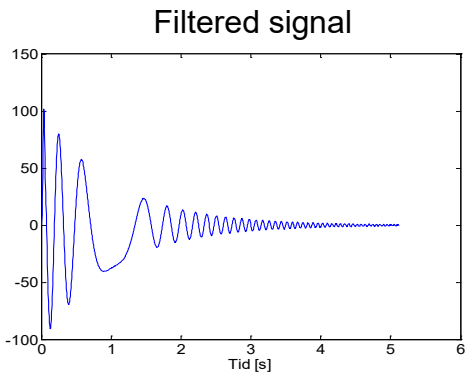
g on f... instance of omega ...

$$g: f(\Omega) \mapsto \mathbb{R}, \qquad \Omega = \{t \mid a < t < b\} \subseteq \mathbb{R}$$
$$y = g \circ f(t) \in \mathbb{R} \qquad t \in (a, b) \subseteq \mathbb{R} \qquad f(t) \in \mathbb{R}$$
$$y = g(f, t) \in \mathbb{R}$$

MOst common way to represent a function

- Example
  - Audio Processing
    - Equalizer (base – middle register – discant)
  - Wireless Communications
    - Radio Transmission





Estimation

- Extracting information, e.g. amplitude, phase, effect, correlation, etc.

What is an Image?

- Image = 2D carrier of visual information, *changing over space*
  - A continuous function: image $f: \Omega \rightarrow \mathbb{R}_+$ $\quad \Omega = \{(x,y) \mid a < x < b,\ c < y < d\} \subseteq \mathbb{R}^2$ $\quad \mathbb{R}_+ = \{x \in \mathbb{R} | x > 0\}$

    $I = f(x,y) \in \mathbb{R}_+$; $\quad (x,y) \in (a,b) \times (c,d) \subseteq \mathbb{R}^2$

  - A space-discrete function: $f: \Omega \rightarrow \mathbb{R}_+$ $\quad \Omega \subseteq \mathbb{Z}^2$
  - A digital image: $f: \Omega \rightarrow \mathbb{Z}_+$ $\quad \Omega \subseteq \mathbb{Z}^2$



What is Image Processing?

Low-pass = smooth/blur.
High-pass = emphasize edges/details.

Båda är exempel på filtering, där vi modifierar en bild för att göra viss information mer eller mindre tydlig.

Filtering

- Accentuating or removing certain aspects
- Described as a function: image $g: f(\Omega) \rightarrow \mathbb{R}_+$ $\Omega = \{(x,y) \mid a < x < b,\ c < y < d\} \subseteq \mathbb{R}^2$
- image $I = g(f,x,y) \subseteq \mathbb{R}_+$, $(x,y) \in \Omega$, $f \in \mathbb{R}_+$

Low Pass filtering
- Smooths sharp edges

High Pass filtering
- Accentuates edges







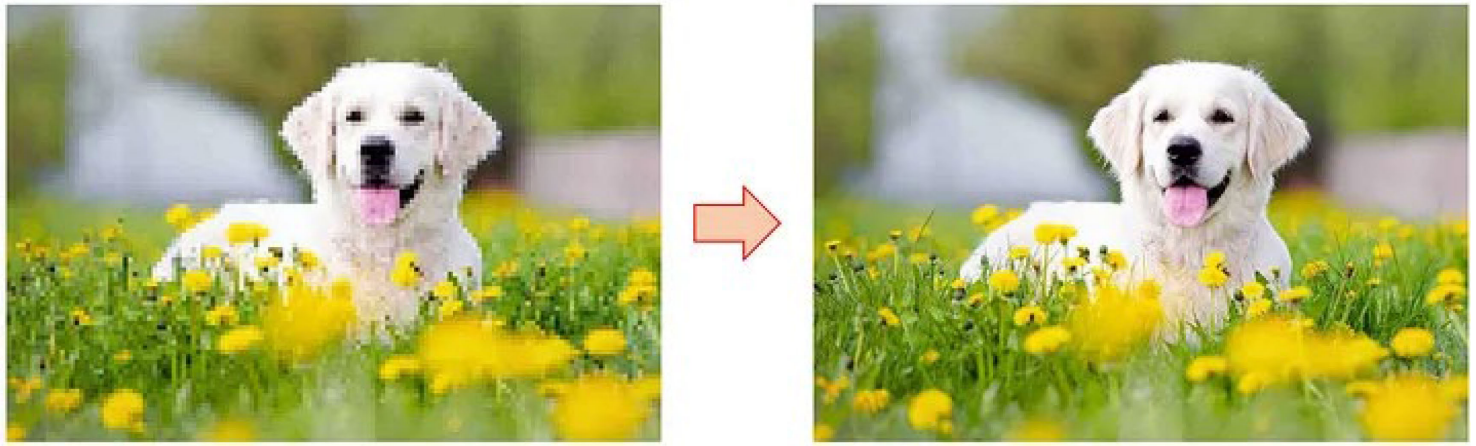
What is Image Processing?

- Inpainting



Jiahui Yu et al., ICCV, 2019

- Spatial super-resolution



https://heartbeat.comet.ml/super-resolution-580424fb7f7c

What is Image Processing?

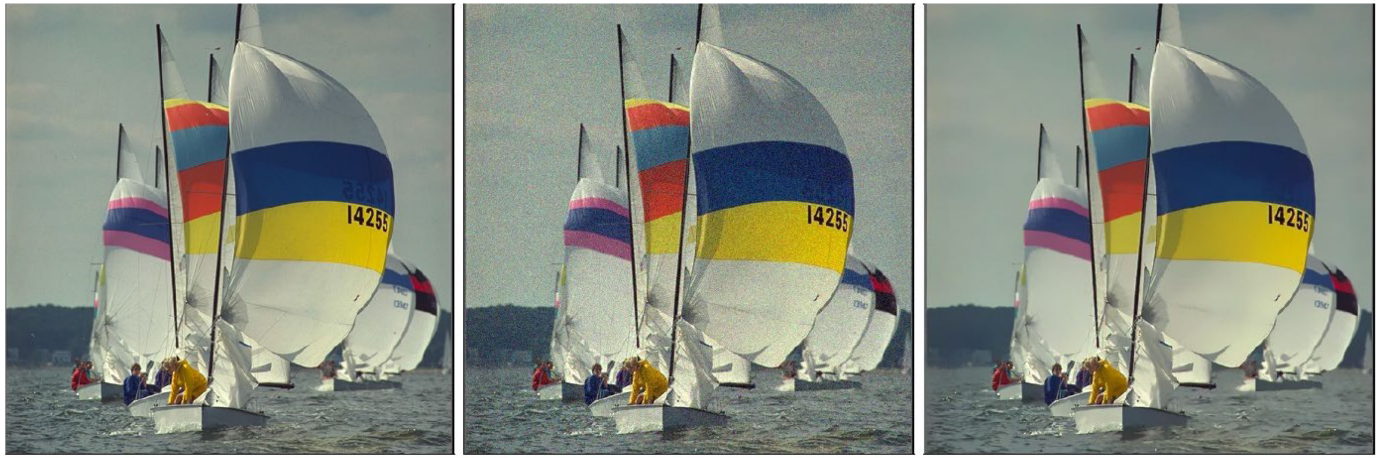
- Deblurring



M Ren, et al., ICCV, 2023

What is Image Processing?

- Denoising



(a) Clean (b) Noisy, $\sigma = 15$ (c) NLM, 33.82dB



(d) BM3D, 36.23dB (e) DnCNN, 36.33dB (f) SwinIR, 37.17dB

M Elad, et al., SIAM J Imaging Sciences 2023

The slide also mentions wavelength w. Real light contains different wavelengths, and wavelength determines colour. So physically, colour can be described using: f(x,y,w)

meaning intensity depends on position (x,y) and wavelength (w).

Processing Visual Data of Higher Dimensions

grayscale is 1d brighness
color images 3d values ... f(x,y) = (R,G,B) ... red, green, blue
f(10,20)=[255,0,0] ... which means that pixel is red.

- Colour Images (3D)
  - Wave-length as input: $f: \Omega \to \mathbb{R}_+$ $\Omega = \{(x, y, \omega) \mid a < x < b,\ c < y < d, \omega\} \subseteq \mathbb{R}^3$

note 3d x,y,w
x has span
y has span
w has small span

  - Red-Green-Blue as output: $f: \Omega \to \mathbb{R}_+^3$ $\Omega = \{(x, y) \mid a < x < b,\ c < y < d\} \subseteq \mathbb{R}^2$

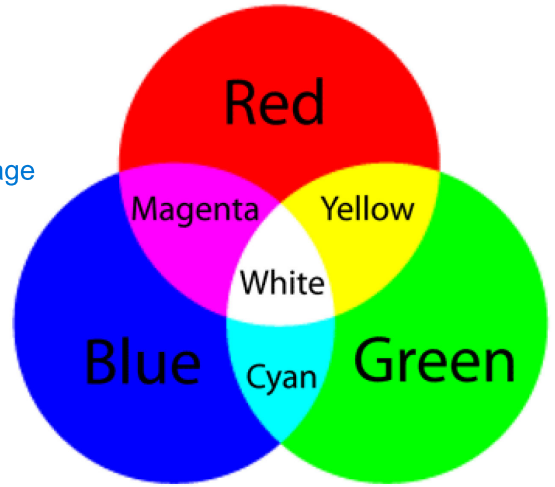
$$\boldsymbol{z} = [R, G, B] = \boldsymbol{f}(x, y) = \boldsymbol{f}(\boldsymbol{x}) \qquad \boldsymbol{x} = (x, y)$$



Rüdiger Katterwe

remember
Grayscale pixel = 1 value
Colour pixel = 3 values } [R,G,B]

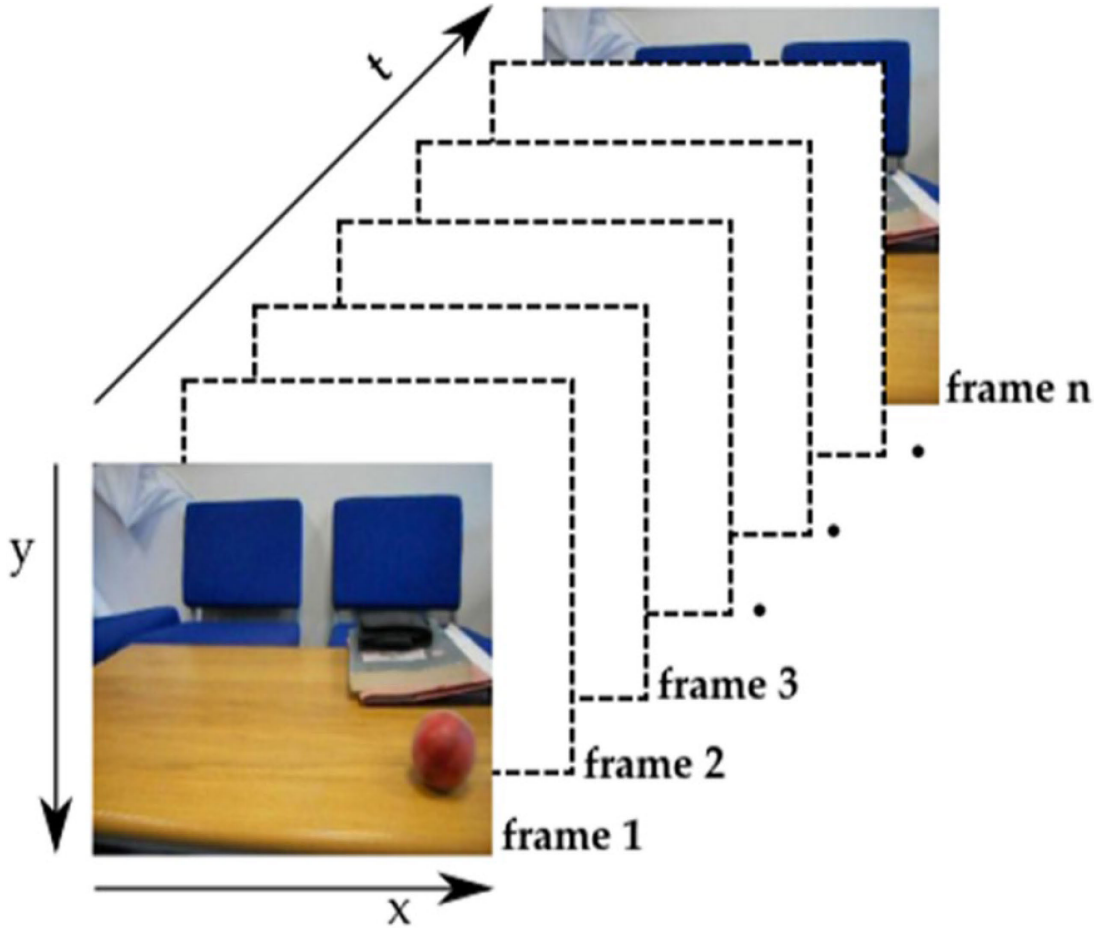
That is why the slide calls colour-image data higher-dimensional vis



https://www.arameb.com/blog/2017/08/31/colors

Processing Visual Data of Higher Dimensions

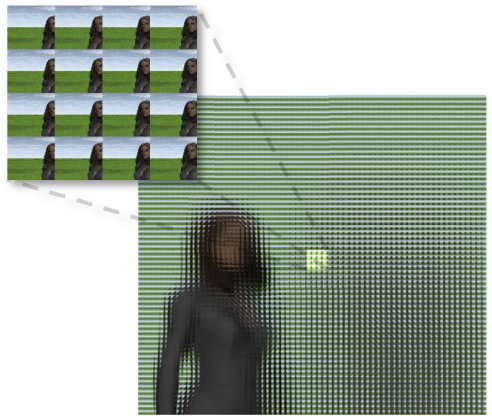
- Video Processing (3D):
  - Filter
  - Inpainting
  - Spatio-temporal super-resolution
  - Deblurring
  - Denoising

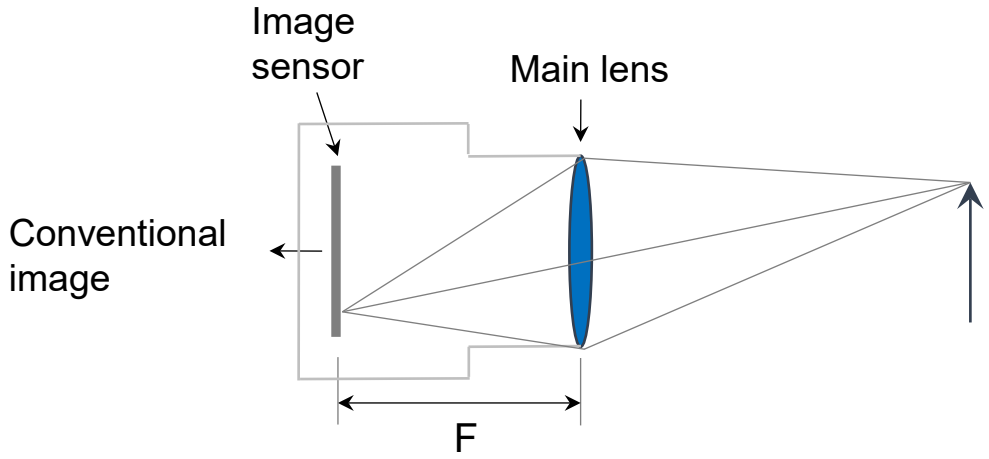


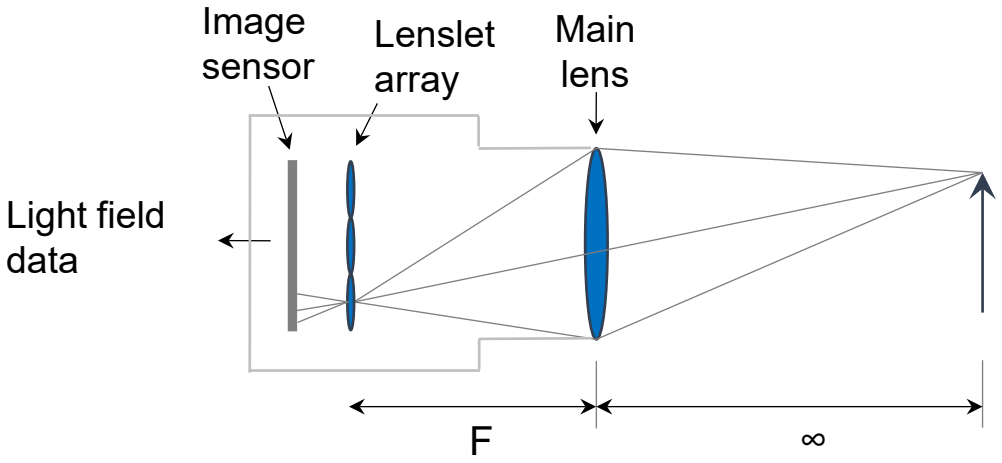
W El-Shafai, et al. Multimedia Tools and Applications. 2023

Processing Visual Data of Higher Dimensions

- Light Field Imaging (4D): Spatial and angular information $I = L(x, y, u, v) \in \mathbb{R}_+$
  - View generation
  - Refocusing
  - Zooming
  - Filtering
  - Inpainting
  - Spatio-angular super-resolution
  - Deblurring
  - Denoising
  - Depth extraction







Processing Visual Data of Higher Dimensions

- Light Field Video (5D): $I = L(x, y, u, v, t) \in \mathbb{R}_+$
  - Same as for LF images
    + spatio-angular-temporal super-resolution

Light field video is basically light field images over time.

So we add \(t\):

I=L(x,y,u,v,t)

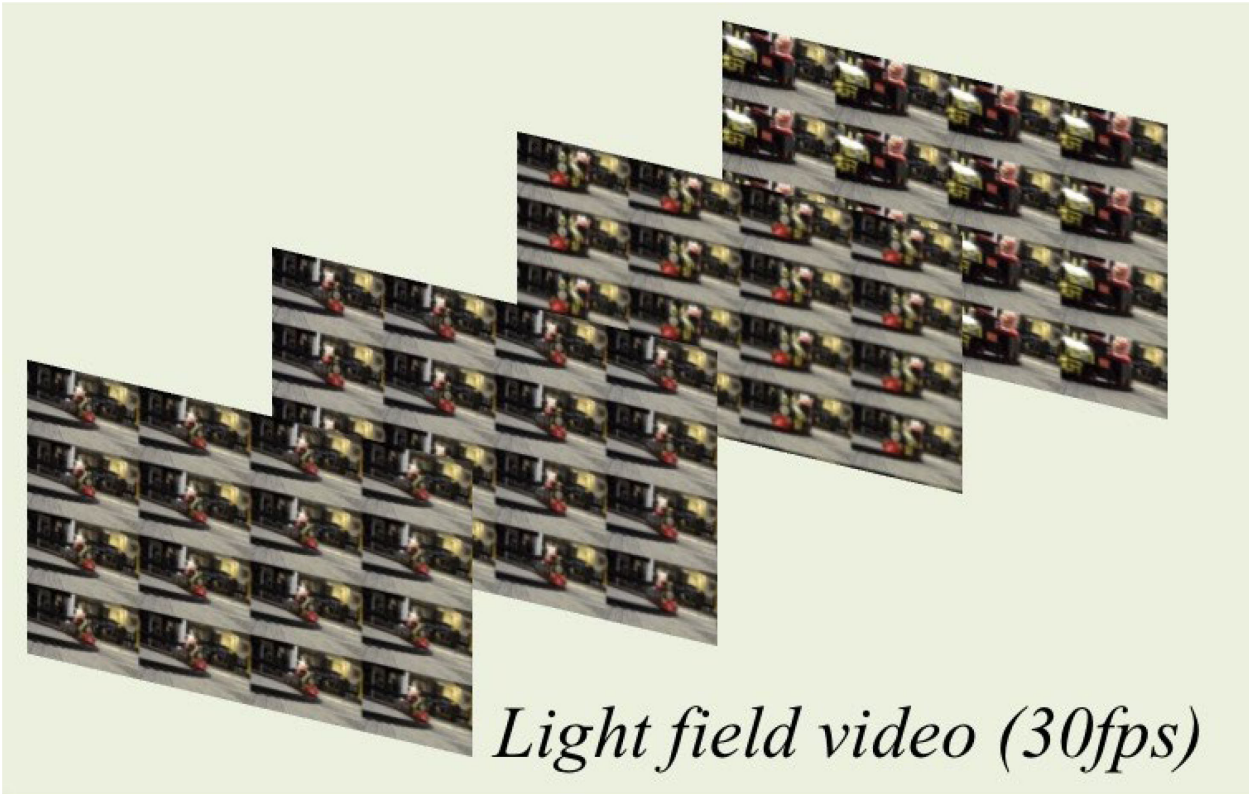
where:

(x,y\) = position
(u,v\) = direction
(t) = time

That makes it 5D.

Think of it as:

4D light-field image + time = 5D light-field video

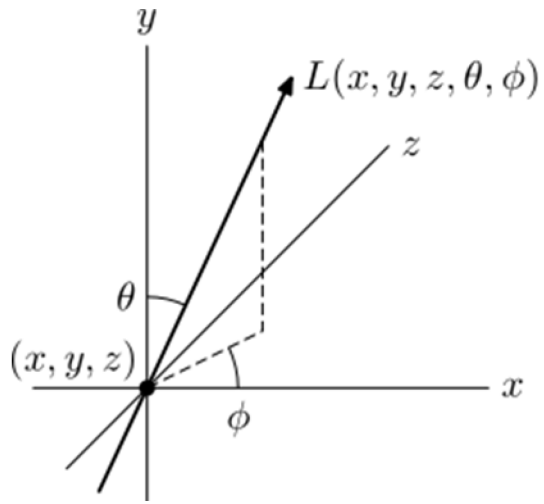


T-C Wang, et al. SIGGRAPH. 2017

Processing Visual Data of Higher Dimensions

- Plenoptic data (7D)
  - Intensity depending on
    - Position (3D)
    - + Direction (2D)
    - + Colour (visual frequency) (1D)
    - + Time (1D)

$$I = P(x, y, z, \phi, \theta, \lambda, t) \in \mathbb{R}_+$$



Fully known plenoptic function → Visual scene could be reproduced precisely

Difference between Image Processing and Computer Vision

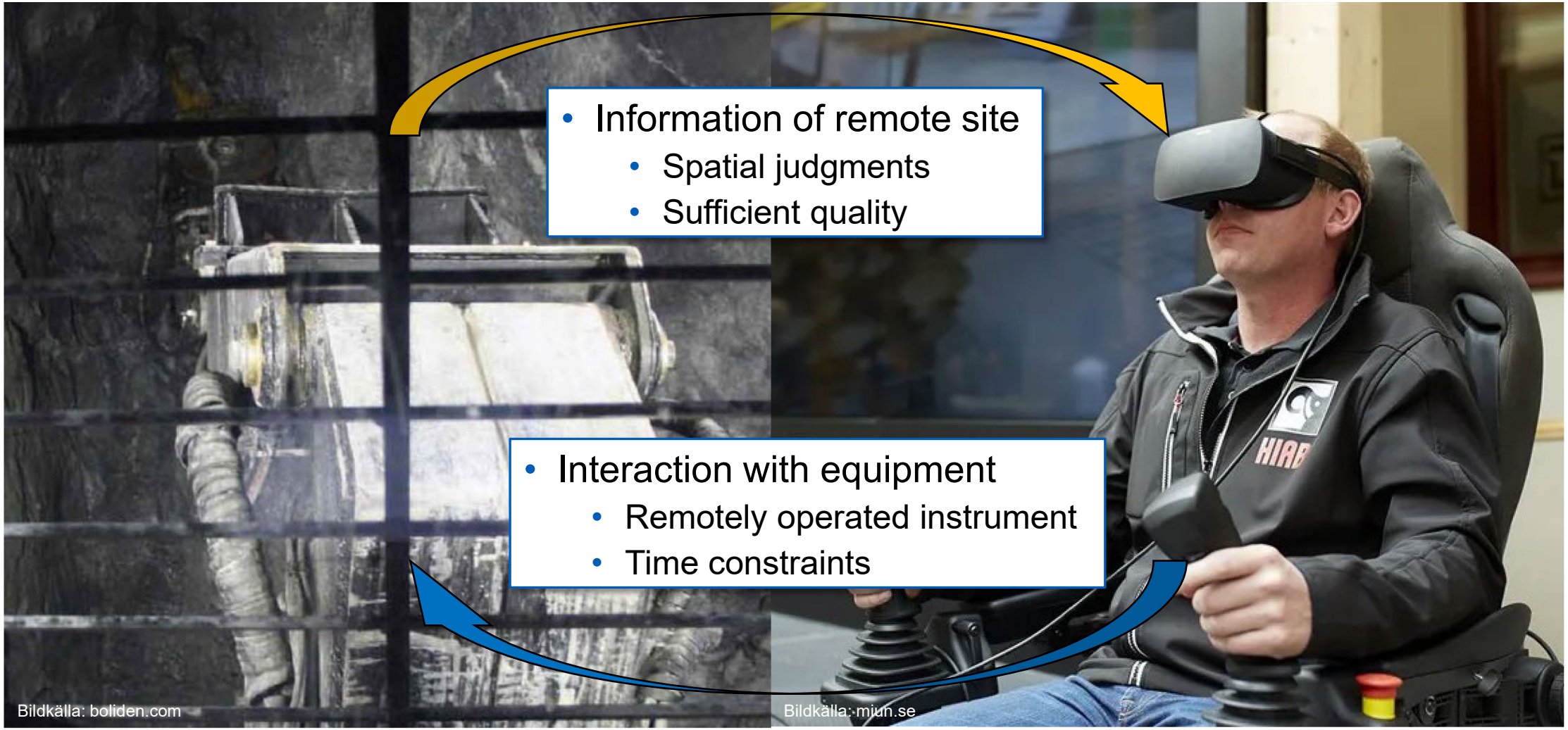
- Image Processing:
  - Image (2D) -> Image (2D)
  - Improved information
- Computer Vision:
  - Image (2D) -> Information (1D or 2D)
  - Examples
    - Classification, Segmentation
    - Positioning
    - Features extraction and matching
    - Two-camera geometry & Depth extraction

- Both support each other
- Classical methods
- Machine and Deep Learning methods

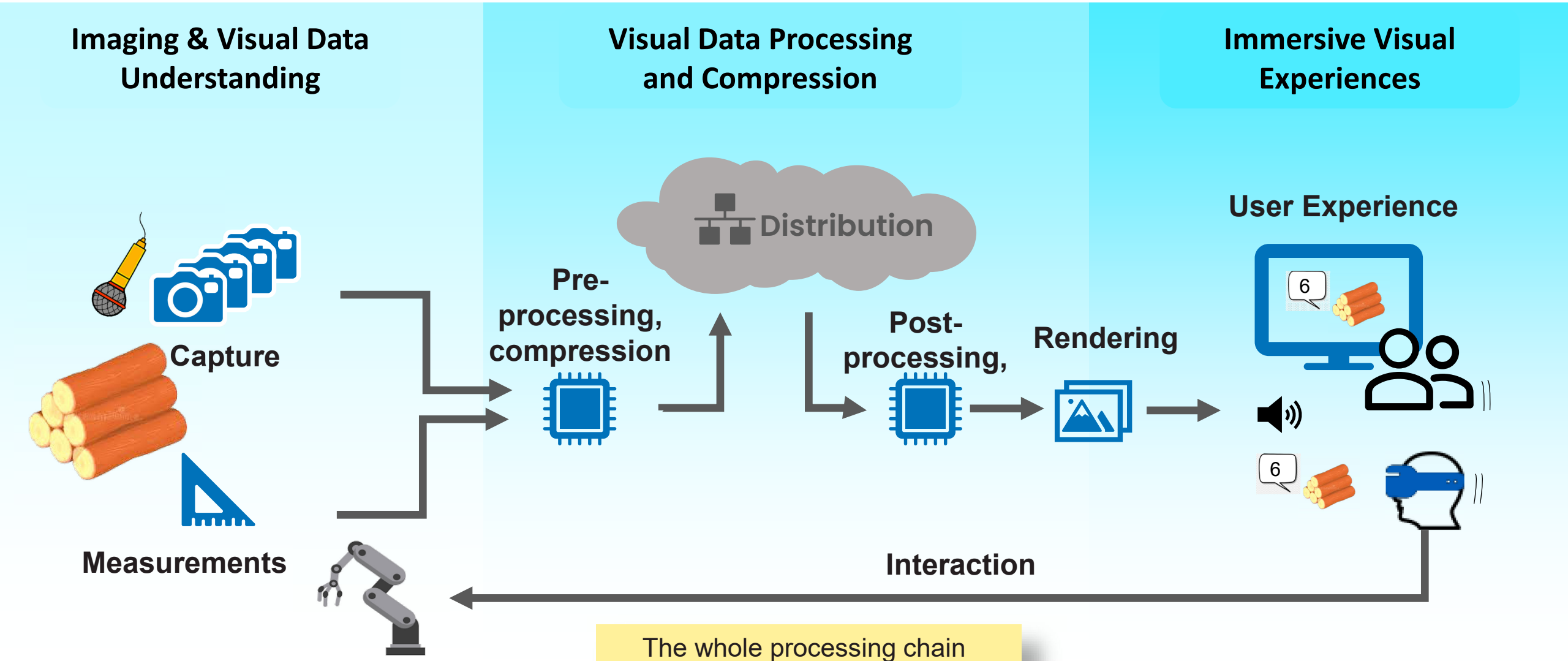
Begränsad delning

Applications

Case Study – Remote Presence and Operation



Case Study – Remote Presence and Operation



The whole processing chain influences the quality of the end-results

Industrial Applications

- First plants in the 1930s
  - Hydroelectric Industry

<u>Renaissance with new technology</u>

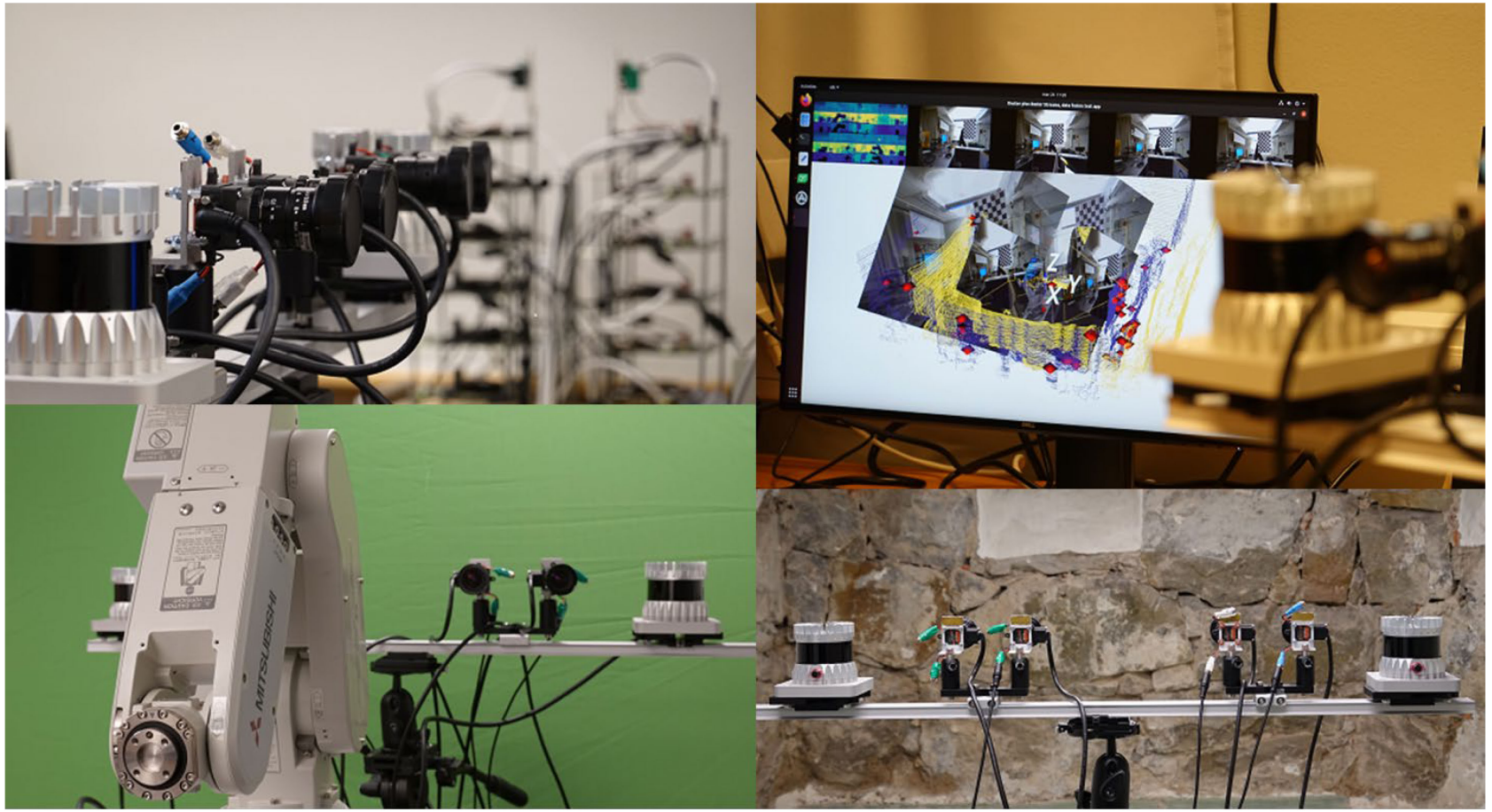
- Airports
- Mining Industry
- Wood Industry
- Crane trucks
- Forestry
- Process Industry
- Medicine and Health care
- Etc.



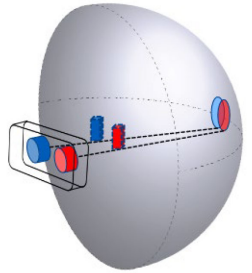
Example – Augmented Space Apprehension

Multiple measurements

- Image
- Distance
- (Sound)
- (Motion – Haptics)
- …

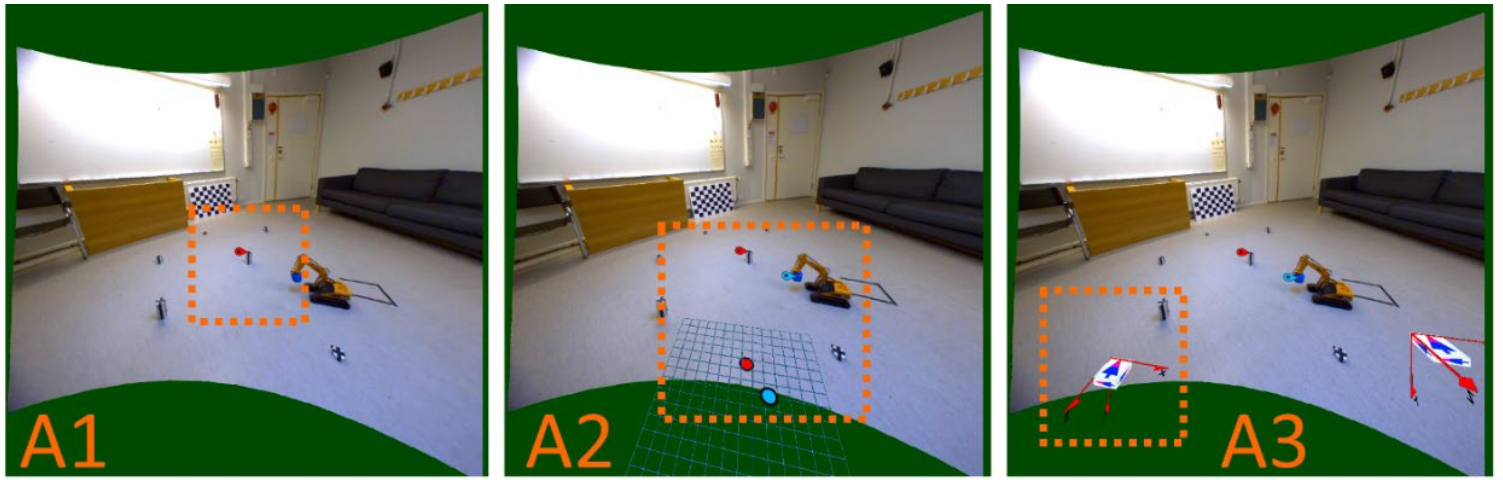


Example – Augmented Space Apprehension



Overlaid Graphics

- Correct Information
- Right placement
- Appropriate form



Example – Augmented Space Apprehension

Extended perspectives

- Virtual views

Spatial filtered perspectives

- Removal of occluding objects



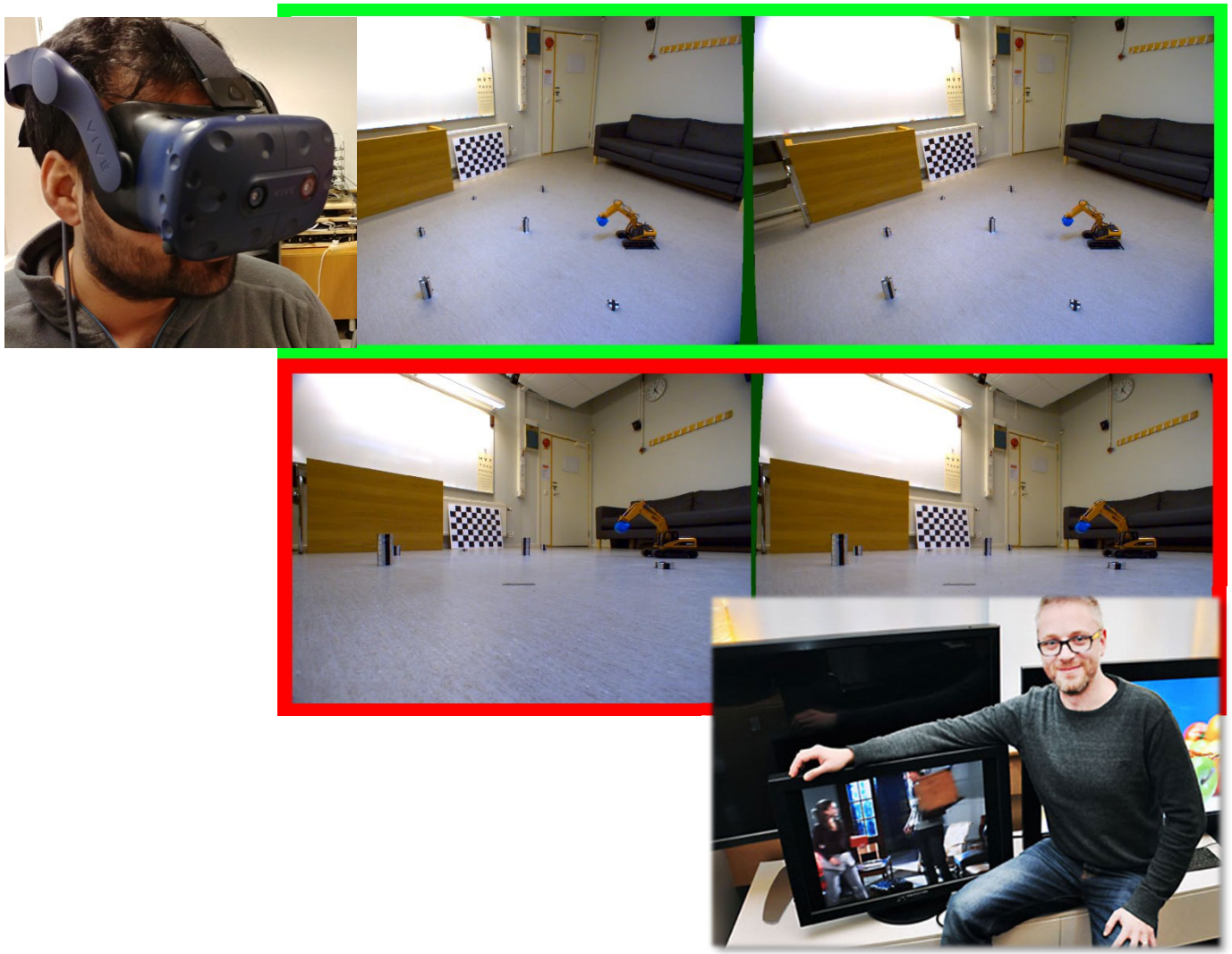
Example – Experienced Quality

<u>Investigations</u>

- Psycho-physiological experiences
- Latency
- Camera positions
- Augmented space apprehension

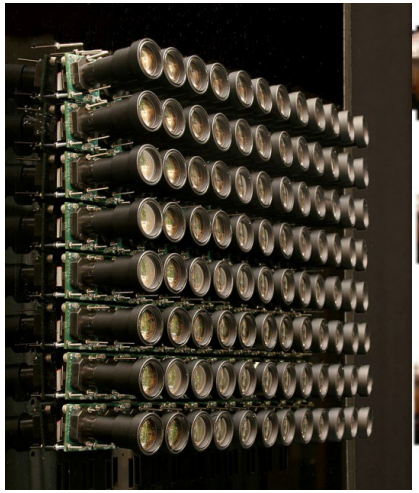
<u>Assessment methodology</u>

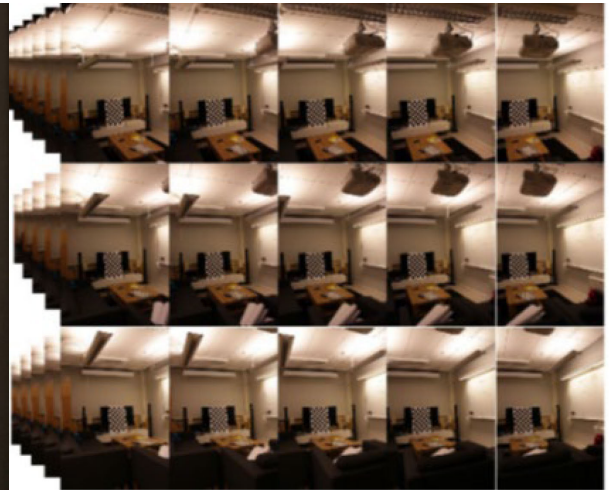
- Performance measures
- Quality of Experience



Example – Video compression

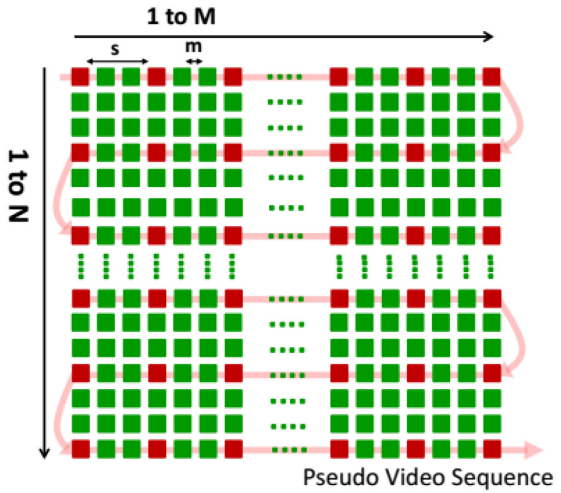
- 2D Video
- Stereo- / Multiview-compression
- Multimodal compression
  - Image + distance + infrared









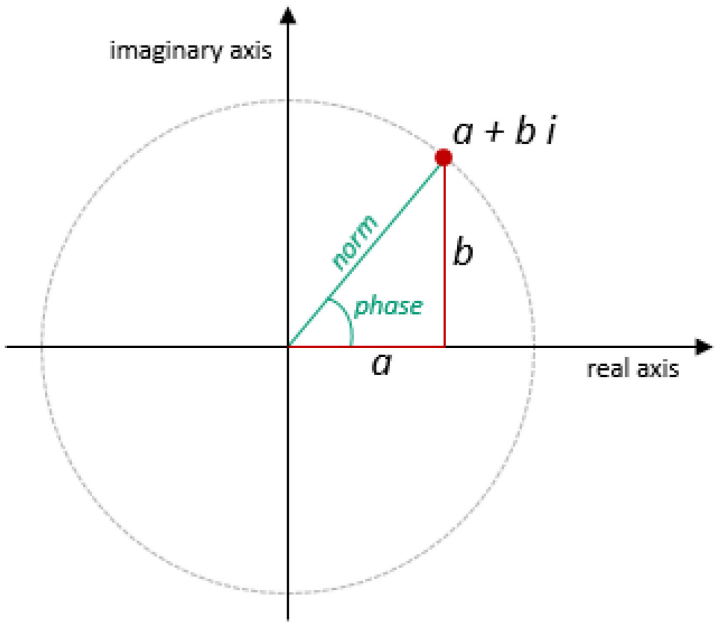


Begränsad delning

Fundamentals

Complex numbers

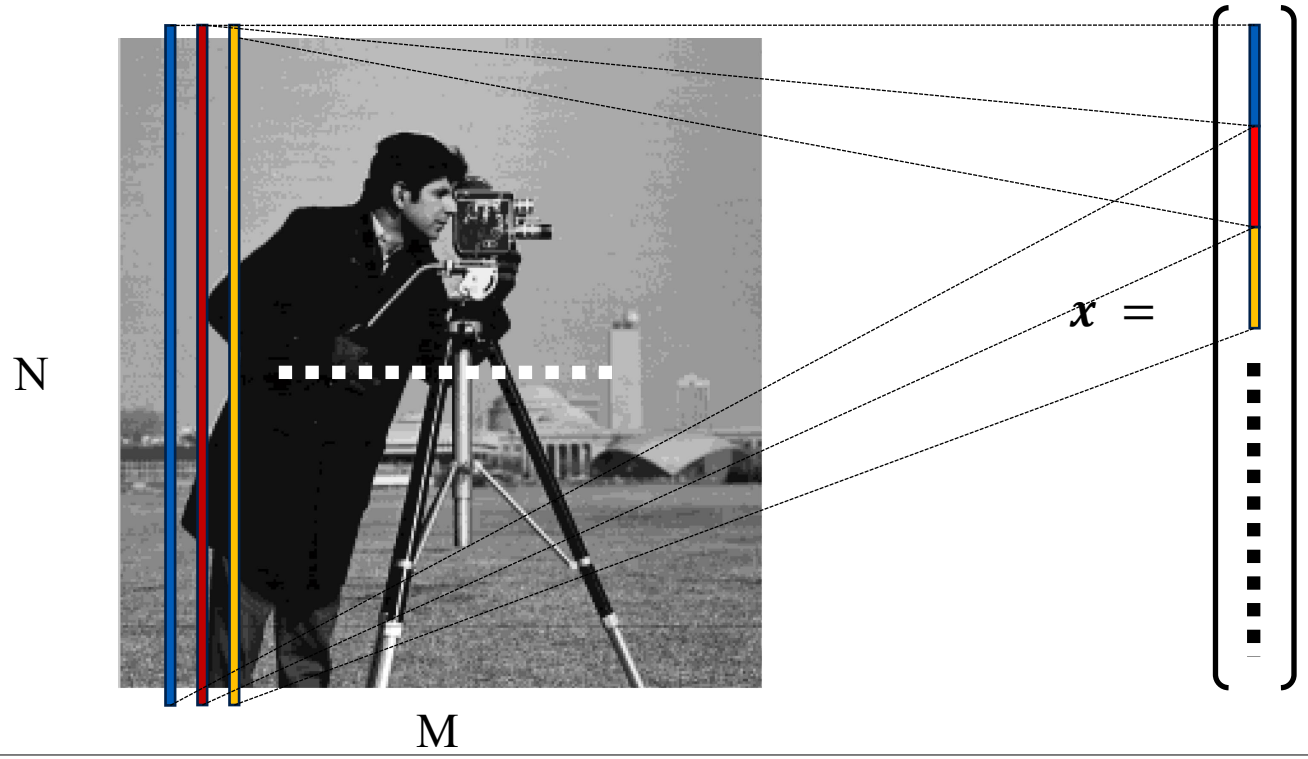
- Imaginary number $j = \sqrt{-1}$
- Complex number $c = a + jb \in \mathbb{C}$
- Complex conjugate $c^* = a - jb$
- Norm / Amplitude $|c|^2 = c \cdot c^* = a^2 + b^2$
- Phase / argument $\varphi = \arg c = \arctan(b/a) + \pi\,(1 - \operatorname{sign} a)$
- Polar form $c = |c|\ \exp(j\varphi)$
- Polar form $c = |c| \cos(\varphi) + j|c| \sin(\varphi)$
- Real part $\mathrm{Re}\,(c) = a = |c| \cos(\varphi)$
- Imaginary part $\mathrm{Im}\,(c) = b = |c| \sin(\varphi)$



Linear Algebra – Vectors

A vector is an ordered list of numbers that represent points, signals, or images

- Ex. a greyscale image $\boldsymbol{x} = [x_1, x_2, \ldots x_n]^T \in \mathbb{R}^{\{n\}}, n = N \times M$
  - The image vector is a concatenation of pixel columns, each entry corresponding to a pixel intensity



Linear Algebra – Matrix Multiplications

- Many operations in Signal and Image Processing are linear operators

  Examples
  - Linear filtering: Convolutions
  - Transforms to more appropriate representations, e.g. frequency domain

- Matrices represent linear operations and transformations

$$\boldsymbol{y} = [y_1, y_2, \dots y_n]^T = \boldsymbol{Ax} \in R^{\{m\}}$$

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} = \left(a_{ij}\right)_{0 \le i \le m-1,\, 0 \le j \le n-1} \in \mathbb{R}^2$$

  - Matrix addition $\boldsymbol{A} + \boldsymbol{B} = \left(a_{ij} + b_{ij}\right)_{0 \le i \le m-1,\, 0 \le j \le n-1}$
  - Scalar multiplication $c \cdot \boldsymbol{A} = \left(c \cdot a_{ij}\right)_{0 \le i \le m-1,\, 0 \le j \le n-1}$
  - Transpose $\boldsymbol{A^T} = \left(a_{ji}\right)_{0 \le j \le n-1, 0 \le i \le m-1}$
  - Matrix multiplication $(\boldsymbol{AB})_{ij} = a_{i0} \cdot b_{0j} + \cdots + a_{i(n-1)} \cdot b_{(n-1)j} = \sum_{r=0}^{n-1} a_{ir} \cdot b_{rj}$

Vector Spaces

- A vector space over a field of scalars $\mathbb{C}$ or $\mathbb{R}$ is a nonempty set V of vectors together with:
  - Vector addition : $(x + y) \in V$
    - $(x + y) = (y + x)$
    - $x + (y + z) = (x + y) + z$
    - $x + 0 = x$
    - $x + (-x) = 0$
  - Scalar multiplication : For every vector $x \in V$ and scalar $\alpha \in \mathbb{C}$, $z = \alpha x \in V$
    - $\alpha(\beta x) = (\alpha\beta)x$
    - $1x = x$
    - $\alpha(x + y) = \alpha x + \alpha y$
    - $(\alpha + \beta)x = \alpha x + \beta x$ where $x, y \in V;\ \alpha, \beta \in \mathbb{C}$
- Span : Given a set of vectors $S = \{x1, x2, .. xr\} \subseteq V$, its span is determined by all linear combinations of those vectors $\text{span}\, S = \{x = \sum_{k=1}^{r} a_k x_k\ ;\ a_k \in \mathbb{C}, x_k \in S\}$
- Linear independence : Vectors $\{x_1, x_2, .. x_r\}$ are linearly independent if $\sum_{k=1}^{r} a_k \mathrm{x}_k = 0$ only when $a_k = 0$ for all k

Vector Spaces

- Finite dimensional space : there exist an integer N such that V contains N linearly independent vectors and any set of $N+1$ vectors is linearly dependent. $N = \dim(V)$
- Infinite dimensional space : no such N exists
- Inner product: is a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{C}$, such that
  $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$
  $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$
  $\langle x, y \rangle = \langle y, x \rangle^*$
  $\langle x, x \rangle \geq 0$, and $\langle x, x \rangle = 0$ iff $x = 0$
- Examples
  - $\mathbb{C}^N$: complex (column) vectors of length N $\quad \langle x, y \rangle = \sum_{k=0}^{N} x_k y_k^* = y^* x$ (c.f. scalar product)
  - $\mathbb{C}^{\mathbb{Z}}$: sequences – discrete time signals $\quad \langle x, y \rangle = \sum_{k \in \mathbb{Z}} x_k y_k^* = y^* x$
  - $\mathbb{C}^{\mathbb{R}}$: functions – continuous time signals $\quad \langle x, y \rangle = \int_{-\infty}^{\infty} x(t) y^*(t) dt$

Norms

- Norm: is a function

$\|\cdot\| : V \to \mathbb{R}$, such that
$\|x\| \geq 0$, and $\|x\| = 0$ iff $x = 0$
$\|\alpha x\| = |\alpha|\|x\|$
$\|x + y\| \leq \|x\| + \|y\|$

- Examples of norms

$\mathbb{C}^N$: $\|x\|_p = \left(\sum_{k=1}^{N} |x_k|^p\right)^{1/p}$; p=2 is Euclidian norm
$\mathbb{C}^{\mathbb{Z}}$: $\|x\|_p = \left(\sum_{k\in\mathbb{Z}} |x_k|^p\right)^{1/p}$; $\|x\|_\infty = \sup|x_k|$ (“max-norm”)
$\mathbb{C}^{\mathbb{Z}}$: $\|x\|_p = \left(\int_{-\infty}^{\infty} |x(t)|^p\, dt\right)^{1/p}$; $\|x\|_\infty = \sup|x(t)|$ (“max-norm”)

- Inner product induces a norm

$$\|x\| = \sqrt{\langle x, x\rangle}$$

- Examples

$\mathbb{C}^N$: $\|x\|_2 = \sqrt{\langle x, x\rangle_2} = \sqrt{\sum_{k=1}^{N} |x_k|^2}$; finite energy N-dim vectors
$\mathbb{C}^{\mathbb{Z}}$: $\|x\|_2 = \sqrt{\langle x, x\rangle_2} = \sqrt{\sum_{k\in\mathbb{Z}} |x_k|^2}$; finite energy discrete-time signals
$\mathbb{C}^{\mathbb{Z}}$: $\|x\|_2 = \sqrt{\langle x, x\rangle_2} = \sqrt{\int_{-\infty}^{\infty} |x(t)|^2 dt}$; finite energy continuous-time signals

Hilbert Spaces

- Hilbert Space is a *complete*, *inner-product*, *normed* space
- Completeness:
  - All vectors can be expressed by the normed vector space.
  - This course only considers complete spaces ☺
- Orthogonality: $\langle x, y \rangle = 0, \quad x \perp y$
- Basis of vector space V: $\Phi = \{\varphi_k\}$ = a set of linearly independent vectors
  - Can uniquely represent any element in V:

$$x = \sum_k^N a_k \varphi_k$$

- Orthogonal basis: all basis vectors are orthogonal

$$\langle \varphi_k, \varphi_n \rangle = 0, k \neq n$$

- Orthonormal basis: : $\langle \varphi_k, \varphi_n \rangle = \delta[k-n]$,
  - Basis vectors are orthogonal and have length 1: $\|\varphi_k\| = 1$
- Coefficients for vector x : $a_k = \frac{\langle x, \varphi_k \rangle}{\langle \varphi_k, \varphi_k \rangle}$, $a_k = \langle x, \varphi_k \rangle$ if $\|\varphi_k\| = 1$
  - I.e. a project of the vector x on the coordinate axis φ_k

Bases – Complex Sinusoids

- Complex exponential function (complex sinusoids)

$$V\colon \Phi = \{\varphi(t) = e^{j\omega t} | \omega \in \mathbb{R}, t \in \mathbb{R}\}, \qquad e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$$

  - Frequency representation

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}dt$$ (inverse Fourier transform)

  - Coefficients

$$X(\omega) = \langle x(t), e^{j\omega t}\rangle = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \in \mathbb{C}$$ (Fourier transform)

Bases - Complex sinusoids for sequences

- Discrete-time complex exponentials functions (complex sinusoids)

$$V\colon \Phi = \{e^{j\Omega n}\},\ \Omega \in \mathbb{R},\ n \in \mathbb{Z} \qquad e^{j\Omega \mathrm{n}} = \cos(\Omega n) + j\sin(\Omega \mathrm{n})$$

  - Frequency representation

$$x(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\Omega})e^{j\Omega \mathrm{n}}d\Omega$$ (inverse Discrete-time Fourier transform)

  - Coefficients

$$X(e^{j\Omega}) = \langle x(n), e^{j\Omega \mathrm{n}}\rangle = \sum_{n=-\infty}^{\infty} x(n)e^{-j\Omega \mathrm{n}} \in \mathbb{C}$$ (Discrete-time Fourier transform)

Bases – Sampled complex sinusoids for sequences

- Discrete complex exponentials functions (complex sinusoids)
  - Consider only N frequency samples around the unit circle:

$$V\colon \Phi = \left\{e^{j\Omega_k n}\right\},\ \Omega_k = \left\{\frac{2\pi k}{N} \middle| 0 \le k \le N-1, k \in \mathbb{Z}\right\} \in \mathbb{R},\ n \in \mathbb{Z}$$

  - Frequency representation

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X\left(e^{j\Omega_k}\right)e^{j\Omega_k} \qquad \text{(inverse Discrete Fourier transform)}$$

  - Coefficients

$$X(k) = X\left(e^{j\Omega_k}\right) = \left\langle x(n), e^{j\Omega_k n}\right\rangle = \sum_{n=0}^{N-1} x(n)e^{-j\Omega_k n} \in \mathbb{C} \qquad \text{(Discrete Fourier transform)}$$

Begränsad delning

Summary

Summary

- Visual AI → AI, Visual sata and Mathematics
- Signal – Carrier of information : $s = f(t) \in \mathbb{R}$ $t \in (a,b) \subseteq \mathbb{R}$
  - Time-continuous, time-discrete, digital
  - Telephony, Sound system
- Signal Processing: $s = g(f,t) \in \mathbb{R}$, $t \in (a,b) \subseteq \mathbb{R}$, $f(t) \in \mathbb{R}$
  - Filtering, estimation
- Image – 2D visual information $I = f(x,y) \in \mathbb{R}_+$, $(x,y) \in (a,b) \times (c,d) \subseteq \mathbb{R}^2$
  - Space-continuous, space-discrete, digital
- Image Processing: $I = g(f,x,y) \subseteq \mathbb{R}_+$, $(x,y) \in \mathbb{R}^2$, $f \in \mathbb{R}_+$
  - Filtering, inpainting, super-resolution
  - Deblurring, Denoising

Summary

- Higher dimensional visual data
  - Colour Images $\boldsymbol{I} = [R, G, B] = \boldsymbol{f}(x, y) = \boldsymbol{f}(\boldsymbol{x})$ $\boldsymbol{x} = (x, y)$
  - Video (3D) $I = f(x, y, t) \in \mathbb{R}_+$
  - Light Field (4D) $I = L(x, y, u, v) \in \mathbb{R}_+$
  - Light Field video (5D) $I = L(x, y, u, v, t) \in \mathbb{R}_+$
  - Plenoptic data (7D) $I = P(x, y, z, \phi, \theta, \lambda, t) \in \mathbb{R}_+$
- Applications
  - Remote presence and operation
    - Airports, Mines, Forestry, Process industry, Medicine and health care
    - Augmented views, AR, Experienced quality, Video compression
- Fundamentals
  - Complex numbers
  - Linear algebra: vectors, matrix multiplications
  - Vector spaces: inner product, norms, Hilbert spaces
  - Fourier Transforms